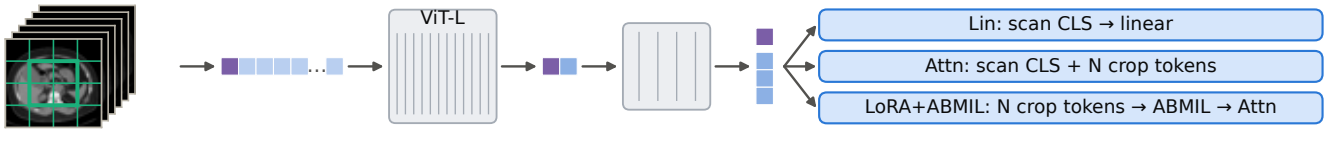
(a) SPECTRE: crops, per-crop ViT-L, feature combiner



512²×256, 0.5×0.5×1 mm, 4³ = 64 crops (train: 2³ = 8)

per 128²×64 crop: 16×16×8 patches, 512 + CLS tokens

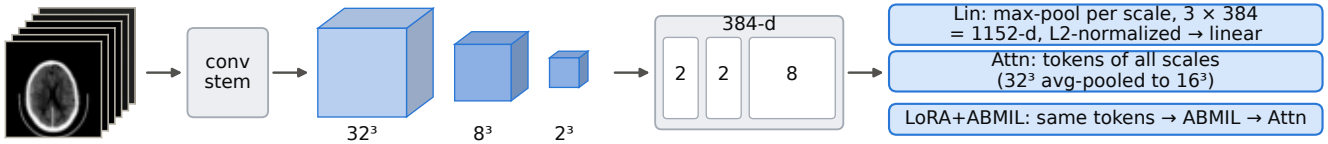
24 blocks, 1080-d, per crop

[CLS ‖ mean], 2160-d

combiner: 4 blocks over crops

scan CLS + N crop tokens

(b) Pillar-0 (head CT): multi-window input, conv stem, three-stage multi-scale encoder



128×256×256, 0.5×0.5×1.25 mm; 11 channels: 10 HU windows + min–max

8×8×4 patches, 32³ grid

tokens at 3 scales (max-pool by 4 per axis), attention within and across scales

3 stages of 2, 2, 8 blocks; scales 32³, 8³, 2³ leave after stages 1, 2, 3; no CLS token

(c) What each classification method trains

		early blocks			late blocks					encoder parameters updated		
	patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE	Pillar-0
Lin								pooled → linear	8000	0	0	0
Attn								attentive probe	8000	0	0	0
LoRA								attentive probe	8000	0.59 M	2.4 M	0.55 M*
LoRA$_{late}$								attentive probe	8000	0.20 M	1.0 M	0.25 M*
LoRA+ABMIL								ABMIL → Attn	8000	0.59 M	2.4 M	0.55 M*
DoRA (+ m)		m			m			attentive probe	8000	0.62 M	2.5 M	0.57 M*
rsLoRA (α/√r)								attentive probe	8000	0.59 M	2.4 M	0.55 M*
Lin→LoRA					2			linear	4000 + 4000	0.20 M	1.0 M	0.25 M*
Attn→LoRA					2			attentive probe	4000 + 4000	0.20 M	1.0 M	0.25 M*
Attn-only FT								attentive probe	8000	28 M	131 M	11 M*
Full FT								attentive probe	8000	87 M	397 M	45 M*

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

W_{qkv}: fused qkv (TAP-CT) or q and kv (SPECTRE, Pillar-0); Attn-only FT trains the whole attention module (with biases).

Late blocks: TAP-CT 9–12, SPECTRE 17–24 and its combiner, the last stage of Pillar-0 (8 of 12 blocks). CVPR26 joint models also ran 26 000 steps (13 000 + 13 000). Head parameters for one label (TAP-CT / SPECTRE / Pillar-0): linear 769 / 1081 / 1153; attentive probe 14.2 / 28.0 / 31.9 M. *Pillar-0: the per-scale blocks of scales that have already left the encoder, and the text projection, are also wrapped or unfrozen but never run and receive no gradient; counting them: LoRA 1.1 M, LoRA$_{late}$ 0.74 M, Attn-only FT 21 M, Full FT 94 M.